

BOSTON UNIVERSITY	TrashformerPro: A Low-Cost Smart Trash Can for Vision-Guided Waste Sorting ME/SE 740 Quad Chart Mid-term Progress Report
1 Project Objective & Scope Changes Since Proposal Design and build a smart trash can that classifies waste items into one of four categories/internal compartments. The waste item is placed on a central plate, which mechanically tilts and dumps based on the visual classification. Changes since proposal: Since the original proposal, the system design has changed based on the project's goals and target users. Instead of mounting the camera and compute hardware above the central plate with a touchscreen for feedback, the camera now points up from below a clear cast acrylic plate. This eliminates the need for the external compute housing and stilts, while reducing the overall footprint of the trash can to roughly that of a small bathroom trash can. Additionally, because a touchscreen on a small trash can would be inconvenient to use, the feedback interface is now a simple phone/web application, on-device LEDs, and buzzers. TrashformerPro System Diagram	2 Research Approach Research tools and objectives: - Public waste-image datasets will be used to bootstrap the vision model. Garbage Dataset (V2) is the main starting dataset, with RealWaste and TrashNet being alternative/additional ones. - Classification categories: Plastic, Paper/Cardboard, Metal/Glass, Trash/Other. - A transfer-learning image classification pipeline will be developed in Python to classify a waste. - The first hardware objective is to test image and classification quality through a clear acrylic plate (with/without linear polarizing film sheet) using an upward facing camera. - A mechanical objective is to design and 3D print a two-axis tilting plate mechanism to dump waste into one of four bins. - The embedded system includes control of the camera, actuators, LEDs, and buzzers. - Later, a phone/web app will be developed and used for user feedback (corrections and fine tuning). Rationale: This research approach emphasizes the main technical hurdles (classification, tilt mechanism, etc.) while also attempting to keep the project cheap and reproducible. The main risk at this moment is getting reliable imaging and classification through the acrylic plate, so that is being tested before finalizing the tilt mechanism.
3 Milestones: Tasks Finished & Tasks Remaining ✓ Tasks Completed - Changed design from an overhead camera to an internal upward facing camera. - Replaced the on-board touchscreen with a web application + on-board LEDs and buzzers. - Identified, downloaded, and inspected initial dataset. - Drafted orthographic sketches with rough dimensions for bin compartments and main housing. - Ordered most hardware components for prototyping and laser-cut acrylic sheet into circular plate. - Created project repo. 🕒 Tasks Remaining - Train a baseline image classification model for the four waste categories - Test imaging and classification through acrylic plate - Design and 3D print plate mount and tilt mechanism - Test actuator control, LEDs, and buzzers 3D print waste bins and main housing - Integrate all components: camera, classifier, tilt mechanism, LED/buzzer feedback	4 Timeline & Final Deliverables Timeline: Final Deliverables: A smart trash can prototype that classifies waste into four internal bins. Equip with a baseline vision model, and onboard LEDs and a buzzer.
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